Code Reference: electronics-search.js

A source-level explanation with function details, file communication, variables, endpoints, and debugging notes.

Item	Explanation
File	electronics-search.js
Purpose	Electronics vocabulary and search support used by the website search behavior.
Lines	162
Size	9,856 bytes
Talks to	GitHub/release layer
Functions	2
Variables/constants	16
API mentions	0
Signals	8

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Rich text at line 8	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",
Security or auth at line 45	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function generator",
Compiler or simulator at line 59	Part of the Compile Code workspace or language/tool detection. Evidence: "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhdl",
Compiler or simulator at line 61	Part of the Compile Code workspace or language/tool detection. Evidence: "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",
Installer at line 80	Windows setup, update, shortcut, or installation behavior. Evidence: "potentiometer", "trimmer", "thermistor", "varistor", "photodiode", "phototransistor", "hall effect sensor",
Cloudflare or AI at line 87	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "layout", "route", "solder", "assemble", "test", "verify", "model", "analyze", "optimize",
Compiler or simulator at line 100	Part of the Compile Code workspace or language/tool detection. Evidence: "ltspice", "spice", "pspice", "ngspice", "matlab", "simulink", "python", "numpy", "scipy",
Git command at line 101	Repository state, publishing, branch checks, or release automation. Evidence: "excel", "kiCad", "kicad", "vivado", "stm32cubeide", "git", "github", "markdown", "c",

Important constants and variables

Item	Explanation
analogTerms	Line 2: const analogTerms = [
powerTerms	Line 20: const powerTerms = [
pcbTerms	Line 33: const pcbTerms = [
testTerms	Line 44: const testTerms = [
digitalTerms	Line 55: const digitalTerms = [
embeddedTerms	Line 66: const embeddedTerms = [
components	Line 76: const components = [

Item	Explanation
actions	Line 85: const actions = [
metrics	Line 92: const metrics = [
toolTerms	Line 99: const toolTerms = [
phraseTerms	Line 105: const phraseTerms = [];
categoryKeywords	Line 114: const categoryKeywords = {
allKeywords	Line 120: const allKeywords = [...new Set([
haystack	Line 139: const haystack = [
buckets	Line 149: const buckets = new Set(categoryKeywords[project?.category] || []);
compactTerm	Line 154: const compactTerm = term.replace(/[^a-z0-9+#]+/g, " ").trim();

JSON/YAML-style keys detected

Item	Explanation
analog-mixed-signal	Line 115: "analog-mixed-signal": [...analogTerms, ...powerTerms, ...pcbTerms, ...testTerms, ...components, ...metrics],

Function-by-function detail

normalize - lines 134-136

Item	Explanation
Source location	Lines 134-136 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalize(value) {
Touches	filesystem
Calls detected	toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
134:   function normalize(value) {
135:     return String(value || "").toLowerCase();
136:   }
```

projectSpecificKeywords - lines 138-158

Item	Explanation
Source location	Lines 138-158 (21 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function projectSpecificKeywords(project, categoryLabel = "") {
Touches	filesystem
Calls detected	map, join, Set, forEach, includes, toLowerCase, add, replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
function projectSpecificKeywords(project, categoryLabel = "") {
  const haystack = [
    project?.title,
    project?.summary,
    project?.category,
    categoryLabel,
    ...(project?.focus || []),
    ...(project?.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title, item.label, item.description].join(" ")),
    ...(project?.languages || [])
  ].map(normalize).join(" ");

  const buckets = new Set(categoryKeywords[project?.category] || []);
  toolTerms.forEach((term) => {
    if (haystack.includes(term.toLowerCase())) buckets.add(term);
  });
  allKeywords.forEach((term) => {
    const compactTerm = term.replace(/[^a-z0-9+#]+/g, " ").trim();
    if (compactTerm && haystack.includes(compactTerm)) buckets.add(term);
  });
  return [...buckets];
}
```

Representative opening snippet

```
(function () {
  const analogTerms = [
    "op amp", "operational amplifier", "fully differential op amp", "differential amplifier", "instrumentation amplifier",
    "transimpedance amplifier", "transconductance amplifier", "current mirror", "cascode", "folded cascode",
    "common source", "common gate", "common drain", "emitter follower", "source follower", "differential pair",
    "active load", "gain stage", "output stage", "input stage", "bias circuit", "bandgap reference",
    "voltage reference", "current reference", "low noise", "offset voltage", "input bias current", "gain bandwidth",
    "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",
    "stability", "compensation", "miller compensation", "frequency response", "bode plot", "pole zero",
    "small signal", "large signal", "noise analysis", "thermal noise", "flicker noise", "shot noise",
    "analog filter", "low pass filter", "high pass filter", "band pass filter", "notch filter", "active filter",
    "rc filter", "rlc circuit", "resonance", "q factor", "cutoff frequency", "impedance", "admittance",
    "matching network", "attenuator", "buffer", "comparator", "window comparator", "schmitt trigger", "hysteresis",
    "oscillator", "vco", "voltage controlled oscillator", "relaxation oscillator", "ring oscillator", "crystal oscillator",
    "pll", "phase locked loop", "charge pump", "sample and hold", "track and hold", "analog switch",
    "multiplexer", "adc", "dac", "data converter", "sigma delta", "successive approximation", "sar adc",
    "pipeline adc", "integrator", "differentiator", "rectifier", "precision rectifier", "peak detector"
  ];

  const powerTerms = [
    "ac to dc", "charger", "battery charger", "power supply", "bench supply", "linear regulator", "ldo",
    "switching regulator", "buck converter", "boost converter", "buck boost", "flyback", "forward converter",
    "sepic", "inverter", "rectifier bridge", "diode bridge", "full wave rectifier", "half wave rectifier",
    "ripple", "load regulation", "line regulation", "transient response", "power stage", "mosfet driver",
    "gate driver", "current limit", "overvoltage", "undervoltage", "reverse polarity", "thermal shutdown",
    "power path", "battery management", "bms", "cc cv", "constant current", "constant voltage", "trickle charge",
    "precharge", "protection circuit", "fuse", "polyfuse", "tvs diode", "esd diode", "snubber", "clamp",
    "soft start", "inrush current", "efficiency", "loss budget", "heat sink", "thermal resistance", "power dissipation",
    "load switch", "ideal diode", "current sense", "shunt resistor", "hall sensor", "isolation", "transformer",
    "optoisolator", "dc dc", "ac dc", "emi filter", "common mode choke", "ferrite bead", "ground loop"
  ];

  const pcbTerms = [
    "pcb", "printed circuit board", "schematic", "layout", "footprint", "symbol", "netlist", "bom",
    "gerber", "drill file", "pick and place", "assembly", "bring up", "board bring up", "prototype",
    "revision", "rev a", "dfm", "dft", "design rule", "clearance", "creepage", "trace width", "via",
    "microvia", "blind via", "buried via", "ground plane", "power plane", "stackup", "controlled impedance",
    "differential pair routing", "return path", "star ground", "analog ground", "digital ground", "guard ring",
    "decoupling capacitor", "bypass capacitor", "bulk capacitor", "test point", "connector", "header",
    "silkscreen", "solder mask", "copper pour", "thermal relief", "stitching via", "keepout", "fiducial",
    "kicad", "altium", "eagle", "orcad", "cadence allegro", "pcbway", "jlcpcb", "osh park"
  ];
```

```

const testTerms = [
  "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function generator",
  "signal generator", "multimeter", "dmm", "electronic load", "power analyzer", "curve tracer", "probe",
  "differential probe", "current probe", "bench test", "validation", "verification", "characterization",
  "measurement", "calibration", "test plan", "test report", "waveform", "transient", "dc sweep", "ac sweep",
  "monte carlo", "corner analysis", "temperature sweep", "load test", "line test", "frequency sweep",
  "gain measurement", "phase measurement", "jitter", "rise time", "fall time", "duty cycle", "pwm",
  "latency", "throughput", "noise floor", "snr", "thd", "thd+n", "eye diagram", "ber", "debug",
  "troubleshooting", "root cause", "failure analysis", "regression test", "unit test", "integration test"
];

const digitalTerms = [
  "digital design", "logic design", "combinational logic", "sequential logic", "finite state machine", "fsm",
  "flip flop", "latch", "counter", "register", "shift register", "clock divider", "clock domain crossing",
  "cdc", "metastability", "synchronizer", "debounce", "timing closure", "setup time", "hold time",
  "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhdl",
  "testbench", "uvm", "simulation", "synthesis", "place and route", "bitstream", "vivado", "quartus",
  "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",
  "apb", "ahb", "spi controller", "i2c controller", "uart", "fifo", "ram", "rom", "bram",
  "pll clock", "mmcm", "serdes", "lvds", "gpio", "interrupt", "dma", "bus protocol"
];

const embeddedTerms = [
  "embedded", "firmware", "microcontroller", "mcu", "stm32", "stm32cubeide", "arm cortex", "cortex m",
  "bare metal", "rtos", "freertos", "scheduler", "interrupt service routine", "isr", "timer",
  "pwm output", "adc input", "dac output", "gpio", "spi", "i2c", "uart", "usb", "can bus",
  "ethernet", "bootloader", "flash memory", "eeprom", "dma", "low power", "sleep mode", "watchdog",
  "sensor interface", "motor control", "control loop", "pid", "encoder", "hall sensor", "imu",
  "temperature sensor", "pressure sensor", "display", "lcd", "oled", "ble", "wifi", "esp32",
```